

# GEORGE JOBI PERANGATTU

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## PROFESSIONAL SUMMARY

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Robotics and Machine Learning Engineer pursuing an M.S. in Robotics and Autonomous Systems (AI) at Arizona State University, with a B.S. in Computer Science. Research experience with PyTorch, multimodal camera-LiDAR learning, GPU-accelerated training, perception, localization, mapping, navigation, model evaluation, simulation, physical robot control, data collection, and failure analysis. Strong software background in Python, C/C++, expert Linux, production ML integration, automated testing, and AI-assisted development workflows.

## EDUCATION

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### M.S. Robotics and Autonomous Systems (AI Concentration) Graduating Jul 2026

Arizona State University, Tempe, AZ

### B.S. Computer Science Graduated May 2025

Arizona State University, Tempe, AZ

## TECHNICAL SKILLS

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**Role Emphasis:** Embodied AI, Physical Intelligence, Vision-Language-Action Concepts, World Model Concepts, Simulation, Sim-to-Real Concepts, Reinforcement Learning Concepts, Imitation Learning Concepts, Multimodal Learning, PyTorch, Data Collection Systems, Training Pipelines, Evaluation Pipelines, Physical Robots, AI-Assisted Coding

**Localization & SLAM:** Robot Localization, Localization Algorithms, SLAM Fundamentals, State Estimation, Kalman Filter Fundamentals, Particle Filter Fundamentals, Sensor Fusion, Camera-LiDAR Fusion, Pose Estimation, Depth Estimation, Mapping, Navigation, Positioning

**Robotic Sensor Systems:** LiDAR, Stereo Cameras, RGB Cameras, RGB-D Cameras, Depth Sensors, Point Cloud Processing, Object Detection, Segmentation, BEV Perception, 3D Perception, Computer Vision, OpenCV, Sensor Integration

**Autonomous Robotics Software:** ROS/ROS 2, Gazebo Simulation, CARLA, URDF, Motion Planning (RRT), Path Planning, Trajectory Optimization, Collision Avoidance, Kinematics, PID Control, Closed-Loop Control, Autonomous Systems

**Languages & Platforms:** C/C++ (Modern), Python, C, Java, JavaScript, TypeScript, SQL, Linux-Based Platforms, Linux/Ubuntu, Embedded Platforms Fundamentals

**ML & Deep Learning:** PyTorch, TensorFlow, Transformers, CNNs, Multi-Task Learning, Mixed-Precision Training, Machine Learning, Data-Driven Analysis, NumPy, Pandas

**Software Engineering:** Git, CMake, CI/CD, Docker, Code Review, Agile, Testing, Validation, Production Code, Documentation, Problem-Solving, GDB, Algorithms, Data Structures

## LOCALIZATION & PERCEPTION PROJECT EXPERIENCE

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### Efficient TransFuser: Camera-LiDAR Sensor Fusion for Localization & Perception Spring 2026

Arizona State University - CSE 598 Research Project

- Emphasized multimodal learning, GPU training, large-scale evaluation, simulation data, failure analysis, and perception-to-action autonomy outputs.
- Developed and optimized robot localization and perception algorithms by designing a camera-LiDAR sensor fusion pipeline, replacing TransFuser's backbone with EfficientNetV2-S to reduce parameters by 70% (168M to 50M) while improving held-out loss by 8.7% and working with LiDAR, stereo cameras, and RGB cameras for autonomous robotic systems.
- Built multi-scale transformer fusion for RGB image and LiDAR point cloud processing, achieving 25.4% BEV segmentation improvement for object detection, pose estimation, depth estimation, and localization in the CARLA autonomous driving simulator.
- Designed data loading and training pipelines for multi-task robotics perception across navigation, depth estimation, semantic segmentation, detection, and BEV mapping, using synthetic data generation and data-driven analysis for model evaluation.
- Ran model evaluations with cosine LR scheduling, gradient clipping, and Kendall uncertainty weighting, writing reusable code and organizing results across multi-task experiments for localization, perception, and navigation outputs.
- Diagnosed an AMP instability in CenterNet's focal loss under FP16 GPU training, documenting, validating, and maintaining high-quality production code with rigorous testing and problem-solving methodology.

## PROJECT EXPERIENCE (CONTINUED)

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### Parrot MiniDrone - Real-Time Localization & Autonomous Control Spring 2026

Arizona State University - Robotics & Autonomous Systems Coursework

- Emphasized physical robot data collection, control, evaluation workflows, and rapid iteration between software and hardware.
- Developed robot localization and navigation algorithms on a physical robotic platform, implementing closed-loop PID control (yaw, velocity, altitude) with sensor fusion of inertial and visual sensor data for real-time state estimation and positioning on Linux-based embedded hardware.
- Built robotics pipelines involving sensor fusion and model inference for automated test sequences, logging sensor data from robotic sensor systems and measuring localization, navigation, and control performance through iterative metrics.
- Engineered safety checks and failsafes ensuring reliable autonomous operation under real-time constraints, demonstrating strong problem-solving skills on real robotic hardware.

### 3D Motion Planning for Autonomous Navigation in Gazebo Fall 2025

Arizona State University - Robotics & Autonomous Systems Coursework

- Developed autonomous robotics software (3D RRT) in Python for collision-free trajectory computation, integrated with Gazebo simulation using ROS for testing, validation, and data-driven analysis of navigation and localization-relevant path planning performance.
- Applied path post-processing to optimize trajectory quality for downstream robotic control, designing experimental details and running evaluations to validate navigation algorithm performance across multiple environment configurations.

### Spider CAD Robot - Multi-Legged Robot Design (In Progress) Spring 2026

Arizona State University - Robotics & Autonomous Systems Coursework

- Designing a multi-legged robot in CAD with constrained joints, validating kinematics and actuator placement; preparing a URDF-exportable model for navigation, localization, and controls experimentation in ROS/Gazebo simulation.

### Factory Automation System for Semiconductor Process Simulation Starting Jun 2026

Arizona State University - Applied Project (Upcoming)

- Will develop an industrial automation system integrating robotic sensor systems, actuators, and basic PLC/control logic for a manufacturing process simulation, applying sensor integration and localization-relevant state tracking across hardware interfaces.

## WORK EXPERIENCE

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### Backend Developer Intern - DigiClips Media Aug 2024 - May 2025

Tempe, AZ

- Developed ML models and data processing pipelines handling 10M+ daily queries using Python and PyTorch, reducing latency by 45% through data-driven analysis, testing, and performance optimization.
- Developed computer vision algorithms for image recognition with cross-functional teams, improving detection/perception accuracy by 30% through reusable code, model evaluation, validation, and iterative problem-solving.

### Full Stack Developer Intern - Odoo May 2025 - Aug 2025

Dubai, UAE

- Refactored Python backend for production applications, contributing reusable infrastructure code and documenting, validating, and maintaining high-quality production code in collaboration with multidisciplinary engineering teams.

### Full Stack Developer Intern - TicketDex Jan 2025 - May 2025

Tempe, AZ

- Delivered core backend services with testing infrastructure, achieving 30% latency reduction through algorithmic optimization, database refactoring, data-driven evaluation, and cross-functional collaboration.